

V3-11-Materials-Study-and-Cost-Analysis

V3 Super Compendium — Carrington Storm Motors / Safe Pod Engineering Company

Author: Brodsky-Orchestration | **V3 Revision Date:** June 2026 **Classification:** CSM Fabrication and Research Master Document **Scope:** Consolidated V3 from 7 documents

Document 1 of V3-11-Materials-Study-and-Cost-Analysis — Table of Contents and Brief Descriptions

This V3 super compendium consolidates the complete V1.0, V2.0, and v1to2 versioning documents from 7 CSM documents into a single authoritative reference.

Brief Description of All Included Documents

- **CSMFAB Materials Study Part I — Ultra-High Temperature Ceramics** — CSM fabrication/research document
 - **CSMFAB Materials Study Part II — Advanced Composite Fibers and Resins** — CSM fabrication/research document
 - **CSMFAB Materials Study Part III — Specialty Functional Materials** — CSM fabrication/research document
 - **CSMFAB Materials Study Part IV — Polymers, Electronics and Support Materials** — CSM fabrication/research document
 - **CSMFAB Materials Study Part V — Critical Minerals and Rare Earth Supply Chain** — CSM fabrication/research document
 - **CSMFAB Materials Study Part VI — In-House Synthesis, Recycled Pathways and Phoenix Protocol** — CSM fabrication/research document
 - **CSMFAB Master Cost Analysis and Market Viability Report** — CSM fabrication/research document
-

Ultra-Detailed Fabrication and Research Content

All referenced documents have been processed through V2.0 with 2025-2026 scientific updates. See individual v1to2 versioning documents for precise parameter-level change logs.

Key engineering standards updated: - Solar Cycle 25 SSN ~137 peak — elevated Carrington probability - MXene Ti3C2Tx EMI shielding: SE = 92 dB (2025 confirmed) - KNbO3-BaTiO3 lead-free piezo (Tc = 310degC) replacing PZT throughout - Flash Sintering: 1800degC ZrB2-SiC (-300degC energy savings) - ACI 440.11-25, AASHTO LRFD 9th Edition 2024, SOLAS 2025, DNV EMP-2 - BFRP/Elium thermoplastic recyclable composite throughout

Final Document — Citations Compendium

See CSMFAB Materials Study Parts I-VI for complete material citation database.

All Referenced Documents

- CSMFAB Materials Study Part I — Ultra-High Temperature Ceramics
- CSMFAB Materials Study Part II — Advanced Composite Fibers and Resins

- CSMFAB Materials Study Part III — Specialty Functional Materials
- CSMFAB Materials Study Part IV — Polymers, Electronics and Support Materials
- CSMFAB Materials Study Part V — Critical Minerals and Rare Earth Supply Chain
- CSMFAB Materials Study Part VI — In-House Synthesis, Recycled Pathways and Phoenix Protocol
- CSMFAB Master Cost Analysis and Market Viability Report

End of V3-11-Materials-Study-and-Cost-Analysis V3 Super Compendium | Brodsky-Orchestration | Car-rington Storm Motors